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LAB EXERCISE 19- CLUSTERIP IN SERVICE IN KUBERNETES

A CLUSTERIP IS THE DEFAULT TYPE OF SERVICE IN KUBERNETES. IT PROVIDES A STABLE INTERNAL IP ADDRESS FOR THE SERVICE, WHICH IS ACCESSIBLE ONLY WITHIN THE CLUSTER. THIS IS USEFUL FOR INTERNAL COMMUNICATION BETWEEN SERVICES WITHIN THE CLUSTER.

STEP 1: CREATE DEPLOYMENT (CORRECT LABELS)

CREATE FILE:

NGINX-DEPLOYMENT.YAML

```
APIVERSION: APPS/V1
KIND: DEPLOYMENT
METADATA:
  NAME: NGINX-APP
SPEC:
  REPLICAS: 2
  SELECTOR:
    MATCHLABELS:
      APP: NGINX # MUST MATCH TEMPLATE LABELS
  TEMPLATE:
    METADATA:
      LABELS:
        APP: NGINX # MUST MATCH SERVICE SELECTOR
    SPEC:
      CONTAINERS:
        - NAME: NGINX
          IMAGE: NGINX
          PORTS:
            - CONTAINERPORT: 80
```

```

GNU nano 7.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx # MUST match service selector
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80

```

SAVE FILE.

KUBECTL APPLY -F NGINX-DEPLOYMENT.YAML

```

ml
deployment.apps/nginx-app created

```

VERIFY:

KUBECTL GET PODS --SHOW-LABELS

NAME	READY	STATUS	RESTARTS	AGE	LABELS
nginx-app-75fdcbbc74-7nhdj	1/1	Running	0	10s	app=nginx, pod-template-hash=75fdcbbc74
nginx-app-75fdcbbc74-9rnm4	1/1	Running	0	10s	app=nginx, pod-template-hash=75fdcbbc74
test-vol1	1/1	Running	0	14m	<none>
test-vol2	1/1	Running	0	12m	<none>

YOU SHOULD SEE:

APP=NGINX

STEP 2: CREATE CLUSTERIP SERVICE (MATCHING SELECTOR)

CREATE FILE:

NGINX-SERVICE.YAML

```

APIVERSION: v1
KIND: SERVICE
METADATA:
  NAME: NGINX-SERVICE
SPEC:
  TYPE: CLUSTERIP
  SELECTOR:
    APP: NGINX # MUST MATCH POD LABEL
  PORTS:
    - PORT: 80
    TARGETPORT: 80

```

```

GNU nano 7.2
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx # MUST match pod label
  ports:
    - port: 80
      targetPort: 80

```

APPLY:

```
KUBECTL APPLY -F NGINX-SERVICE.YAML
```

STEP 3: VERIFY IT IS WORKING

CHECK SERVICE

```
KUBECTL GET SVC
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	18d
nginx-service	ClusterIP	10.97.52.103	<none>	80/TCP	12s
nodeport-service	NodePort	10.101.115.216	<none>	80:30007/TCP	18d

YOU SHOULD SEE:

```

TYPE      CLUSTER-IP
CLUSTERIP 10.X.X.X

```

MOST IMPORTANT CHECK

```
KUBECTL GET ENDPOINTS NGINX-SERVICE
```

```

Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS                                AGE
nginx-service  10.244.0.100:80,10.244.0.99:80      31s

```

NOW YOU SHOULD SEE SOMETHING LIKE:

```
10.244.x.x:80, 10.244.x.x:80
```

STEP 4: TEST FROM INSIDE CLUSTER

RUN TEST POD:

```
KUBECTL RUN TEST-POD --IMAGE=BUSYBOX --RM -IT -- SH
```

INSIDE POD:

```
WGET -qO- HTTP://NGINX-SERVICE
```

```
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # |
```

YOU SHOULD SEE:

WELCOME TO NGINX!

EXIT:

EXIT

STEP 5: TEST FROM OUTSIDE (IT WILL FAIL)

FROM YOUR LOCAL MACHINE:

CURL HTTP://<CLUSTER-IP>

IT WILL NOT WORK.

THAT PROVES:

CLUSTERIP WORKS ONLY INSIDE CLUSTER.